



Zynq 7000 SoC and 7 Series Devices Memory Interface Solutions v4.2, LogiCORE IP Data Sheet

DS176 (v4.2) November 13, 2024

Product Specification

Introduction

The AMD Zynq™ 7000 SoC and 7 series FPGAs memory interface solutions cores provide high-performance connections to [DDR3 and DDR2 SDRAMs](#), [QDR II+ SRAM](#), [RLDRAM II/RLDRAM 3](#), and [LPDDR2 SDRAM](#).

IP Facts

AMD LogiCORE™ IP Facts Table	
Core Specifics	
Supported Device Family ¹	Zynq 7000 SoC, 7 series ² FPGAs
Supported Memory	DDR3 Component and DIMM, DDR2 Component and DIMM, QDR II+, RLDRAM II, RLDRAM 3, and LPDDR2 SDRAM Components
Resources	See Resource Utilization
Provided with Core	
Documentation	Product Specification User Guide
Design Files	Verilog, VHDL (top-level files only)
Example Design	Verilog, VHDL (top-level files only)
Test Bench	N/A
Constraints File	XDC
Supported S/W Driver	N/A
Tested Design Flows ³	
Design Entry	AMD Vivado™ Design Suite
Simulation	For supported simulators, see the <i>Vivado Design Suite User Guide: Release Notes, Installation, and Licensing</i> (UG973).
Synthesis ⁴	Vivado Synthesis
Support	
Release Notes and Known Issues	Master Answer Record: 75889
All Vivado IP Change Logs	Master Vivado IP Change Logs: 72775
Support web page	

Notes:

- For a complete listing of supported devices, see the release notes for MIG.
- See the *Virtex 7 FPGAs Data Sheet: DC and AC Switching Characteristics* ([DS183](#)), the *Kintex 7 FPGAs Data Sheet: DC and AC Switching Characteristics* ([DS182](#)), or *Spartan 7 FPGAs Data Sheet: DC and AC Switching Characteristics* ([DS189](#)) for performance information.
- For the supported versions of the tools, see the *Vivado Design Suite User Guide: Release Notes, Installation, and Licensing* ([UG973](#)).
- The standard synthesis flow for Synplify is not supported for the core.

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DDR3 and DDR2 SDRAMs

This section discusses the features, applications, and functional description of the 7 series FPGAs memory interface solutions in DDR3 and DDR2 SDRAMs. These solutions are available with an optional AXI4 slave interface.

DDR3 SDRAM Features

- Component support for interface widths up to 72 bits
- Single and dual rank UDIMM, RDIMM, and SODIMM support
- DDR3 (1.5V) and DDR3L (1.35V)
- 1, 2, 4, and 8 Gb density device support
- 8-bank support
- x8 and x16 device support
- 8:1 DQ:DQS ratio support
- Configurable data bus widths (multiples of 8, up to 72 bits)
- 8-word burst support
- Support for 5 to 14 cycles of column-address strobe (CAS) latency (CL)
- On-die termination (ODT) support
- Support for 5 to 10 cycles of CAS write latency
- ZQ calibration – initial and periodic (configurable)
- Write leveling support for DDR3 (fly-by routing topology required for DDR3 component designs)
- JEDEC®-compliant DDR3 initialization support
- Source code delivery in Verilog and VHDL (top-level files only)
- 4:1 and 2:1 memory to FPGA logic interface clock ratio
- ECC support
- I/O Power Reduction option reduces average I/O power by automatically disabling DQ/DQS IBUFs and internal terminations during writes and periods of inactivity
- Internal V_{REF} support
- Multicontroller support for up to eight controllers
- Two controller request processing modes:
 - Normal: reorder requests to optimize system throughput and latency
 - Strict: memory requests are processed in the order received

DDR2 SDRAM Features

- Component support for interface widths up to 64 bits

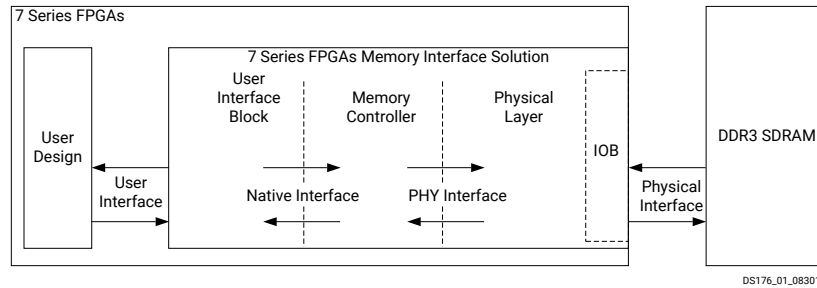
- Single rank UDIMM, RDIMM, and SODIMM
- 1 and 2 Gb density device support (additional densities supported in the MIG tool using the Create Custom Part feature)
- 4- and 8-bank support
- x8 and x16 device support
- 8:1 DQ:DQS ratio support
- Configurable data bus widths (multiples of 8, up to 72 bits)
- 8-word burst support
- Support for 3 to 6 cycles of column address strobe (CAS) latency
- On-die termination (ODT) support
- JEDEC-compliant DDR2 initialization support
- Source code delivery in Verilog and VHDL (top-level files only)
- 4:1 and 2:1 memory to FPGA logic interface clock ratio
- ECC support
- I/O Power Reduction option reduces average I/O power by automatically disabling DQ/DQS IBUFs and internal terminations during writes and periods of inactivity
- Internal V_{REF} support
- Two controller request processing modes:
 - Normal: Reorder requests to optimize system throughput and latency
 - Strict: Memory requests are processed in the order received
- Multiple controllers per FPGA supported running the MIG tool multiple times

Applications

Typical applications for the 7 series FPGAs memory interface solutions include DDR3 SDRAM and DDR2 SDRAM interfaces.

The following figure shows a high-level block diagram of the 7 series FPGAs memory interface solution connecting a user design to a DDR2 or DDR3 SDRAM device. The physical layer (PHY) side of the design is connected to the DDR2 or DDR3 SDRAM device through FPGA I/O blocks (IOBs), and the user interface side is connected to the user design through FPGA logic. For more details regarding the design, see the *Zynq 7000 SoC and 7 series Devices Memory Interface Solutions* ([UG586](#)).

Figure 1: DDR2/DDR3 SDRAM Memory Interface Solution



Functional Description

As shown in the figure, the top-level functional blocks of the 7 series FPGAs memory interface solution include:

- The User Interface block:
 - Presents the user interface to a user design
 - Provides a simple and user-friendly alternative to the native interface
 - Buffers read and write data
 - Reorders read return data to match the request order
 - Presents a flat address space and translates it to the addressing required by the SDRAM
- The Memory Controller block:
 - Receives requests from the user design
 - Reorders requests to minimize dead states for maximum SDRAM performance
 - Manages SDRAM row/bank configuration
 - Performs high-level SDRAM management such as refresh and activate/precharge
- The PHY block:
 - Interfaces with the Memory Controller block over a simple interface and translates the signals into the actual signals sent to the SDRAM, and vice versa
 - Translates and synchronizes control and data over various clock domains
 - Initializes the SDRAM
 - Performs write leveling for DDR3 (fly-by routing topology required for component designs)
 - Performs calibration to center align capture clocks with read data

The figure also shows a user design connecting to the memory interface. An example user design is provided with the core. For more details regarding the design, see the *Zynq 7000 SoC and 7 series Devices Memory Interface Solutions* ([UG586](#)).

AXI4 Slave Interface Features

These features are optional and selectable using the MIG GUI:

- AMBA® AXI4 slave-compliant memory-mapped interface
- AXI4-Lite interface support for ECC control and status registers
- 1:1 clock rate to the controller
- AXI4 interface data widths can be 64, 128, 256, or 512 bits to correspond with memory data widths of 8, 16, 32, 64, or 72 bits (72 bits is supported when 64 bits of data and 8 bits of ECC is used)
- Parameterized address width support
- Support for incremental (INCR) burst up to 256 data beats
- WRAP burst support
- Multicontroller support for up to eight DDR3 SDRAM controllers

QDR II+ SRAM

This section discusses the features, applications, and functional description of the 7 series FPGAs memory interface solutions in QDR II+ SRAMs.

Features

- QDR II+ SRAM device support
- x18 and x36 memory width support
- 36 Mb, 72 Mb, and 144 Mb density device support
- Configurable data bus widths (x18, x36)
- 2-word and 4-word burst support
- Source code delivery in Verilog and VHDL (top-level files only)
- 2:1 memory to FPGA logic interface clock ratio
- 2.0-cycle and 2.5-cycle read latency support
- Fixed latency mode support
- Internal V_{REF} support
- Multicontroller support for up to eight controllers

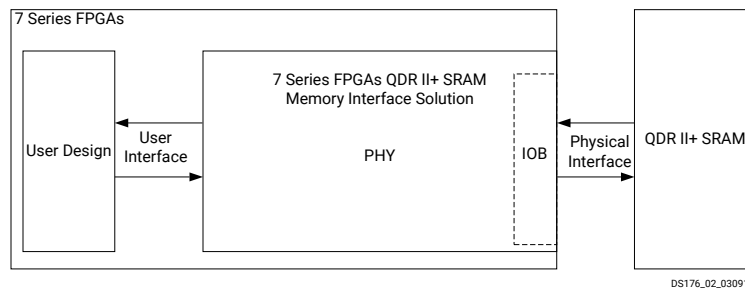
Applications

QDR II+ SRAMs offer high-speed data transfers on separate read and write buses on the rising and falling edges of the clock. These memory devices are used in high-performance systems as temporary data storage, such as:

- Look-up tables in networking systems
- Packet buffers in network switches
- Cache memory in high-speed computing
- Data buffers in high-performance testers

The following figure shows a high-level block diagram of the 7 series FPGAs memory interface solution connecting a user design to a QDR II+ SRAM device.

Figure 2: QDR II+ SRAM Memory Interface Core



Functional Description

As shown in the figure, the top-level functional block is composed of a PHY that interfaces to the user and to the QDR II+ SRAM device. The PHY block:

- Translates simple user read and write commands to conform to QDR II+ SRAM protocol
- Enables the user to provide up to one read and one write transaction per clock cycle for maximum throughput
- Performs calibration to center align clocks with data
- Returns data to the user with a corresponding valid signal
- Translates and synchronizes over various clock domains
- Implements an optimized half-frequency design that eliminates the need for a memory controller

For more details regarding the design, see the *Zynq 7000 SoC and 7 series Devices Memory Interface Solutions* (UG586) provided with the core.

RLDRAM II/RLDRAM 3

This section discusses the features, applications, and functional description of the 7 series FPGAs memory interface solutions in RLDRAM II and RLDRAM 3 devices.

RLDRAM II Features

- RLDRAM II common I/O (CIO) memory device support
- x18 and x36 memory width support
- 288 Mb and 576 Mb density device support
- Configurable data bus widths (x18, x36, x72)
- 4-word and 8-word burst support
- Configuration 1, 2, 3 support
- Address Multiplexing Mode support

- ODT support
- Source code delivery in Verilog and VHDL (top-level files only)
- 2:1 memory to FPGA logic interface clock ratio
- Internal V_{REF} support
- Multicontroller support for up to eight controllers

RLDRAM 3 Features

- x18 and x36 memory width support
- 576 Mb and 1152 Mb density device support
- Configurable data bus widths (x18, x36, x72)
- 2-word, 4-word, and 8-word burst support
- Address Multiplexing Mode support
- ODT support
- Source code delivery in Verilog only
- 4:1 memory to FPGA logic interface clock ratio
- Internal V_{REF} support

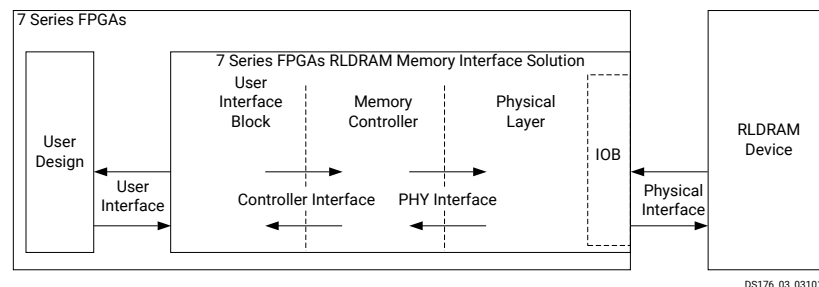
Applications

RLDRAM II and RLDRAM 3 devices are used in high-performance systems as temporary data storage, such as:

- Look-up tables in networking systems
- Packet buffers in network switches
- Cache memory in high-speed computing
- Data buffers in high-performance testers

The following figure shows a high-level block diagram of the 7 series FPGAs memory interface solution connecting a user design to an RLDRAM device. The physical layer is connected to the RLDRAM device through FPGA IOBs, and the user interface is connected to the user design through FPGA logic.

Figure 3: RLDRAM Memory Interface Core



Functional Description

As shown in the figure, the top-level functional blocks of the RLDRAM memory interface solution include:

- The User Interface block:
 - Presents the user interface to a user design
 - Buffers commands and write data
- The Memory Controller block:
 - Receives requests from the user design
 - Processes commands in order and adheres to memory specifications
 - Performs high-level SDRAM management, such as refresh, and controls bank access
- The Physical Layer (PHY) block:
 - Interfaces with the Memory Controller block over a simple interface and translates the signals into the actual signals sent to the RLDRAM, and vice versa.
 - Performs memory initialization sequence.
 - Performs calibration to center align clocks with data
 - Returns data to the user with a corresponding valid signal

For more details regarding the design, see the *Zynq 7000 SoC and 7 series Devices Memory Interface Solutions (UG586)* provided with the core.

LPDDR2 SDRAM

This section discusses the features, applications, and functional description of the 7 series FPGAs memory interface solutions in LPDDR2 SDRAMs.

Features

- Component support for interface widths up to 32 bits
- 2 and 4 Gb density device support
- 8-bank support
- x16 and x32 device support
- 8:1 DQ:DQS ratio support
- 8-word burst support
- JEDEC-compliant LPDDR2 SDRAM initialization support
- Source code delivery in Verilog
- 2:1 memory to FPGA logic interface clock ratio
- Internal V_{REF} support

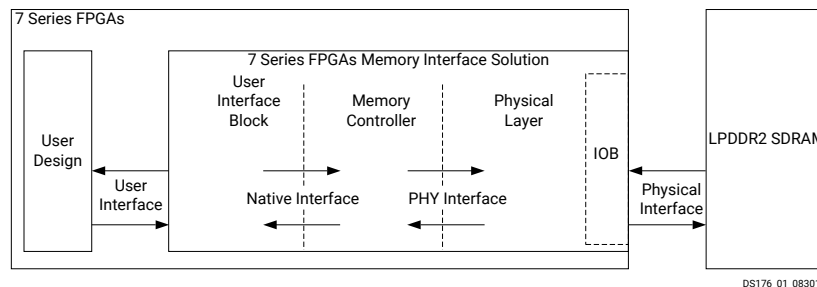
- Two controller request processing modes:
 - Normal: reorder requests to optimize system throughput and latency
 - Strict: memory requests are processed in the order received

Applications

Typical applications for the 7 series FPGAs memory interface solutions include LPDDR2 SDRAM interfaces.

The following figure shows a high-level block diagram of the 7 series FPGAs memory interface solution connecting a user design to a LPDDR2 SDRAM device. The physical layer (PHY) side of the design is connected to the LPDDR2 SDRAM device through FPGA I/O blocks (IOBs), and the user interface side is connected to the user design through FPGA logic. For more details regarding the design, see the *Zynq 7000 SoC and 7 series Devices Memory Interface Solutions* ([UG586](#)) provided with the core.

Figure 4: LPDDR2 SDRAM Memory Interface Solution



Functional Description

As shown in the figure, the top-level functional blocks of the 7 series FPGAs memory interface solution include:

- The User Interface block:
 - Presents the user interface to a user design
 - Provides a simple and user-friendly alternative to the native interface
 - Buffers read and write data
 - Reorders read return data to match the request order
 - Presents a flat address space and translates it to the addressing required by the SDRAM
- The Memory Controller block:
 - Receives requests from the user design
 - Reorders requests to minimize dead states for maximum SDRAM performance
 - Manages SDRAM row/bank configuration
 - Performs high-level SDRAM management such as refresh and activate/precharge

- The Physical Layer (PHY) block:
 - Interfaces with the Memory Controller block over a simple interface and translates the signals into the actual signals sent to the SDRAM, and vice versa.
 - Translates and synchronizes control and data over various clock domains
 - Initializes the SDRAM
 - Performs calibration to center align capture clocks with read data

The figure also shows a user design connecting to the memory interface. For more details regarding the design, see the *Zynq 7000 SoC and 7 series Devices Memory Interface Solutions* ([UG586](#)) provided with the core.

General Specifications

For more details regarding specific banking, pin location, and internal clock resource requirements for all cores, see the *Zynq 7000 SoC and 7 series Devices Memory Interface Solutions* ([UG586](#)).

Resource Utilization

Table 1: Resource Utilization for 7 Series FPGAs

Product ¹	LUTs	Flip-Flops	BUFG	PLLE2	MMCM	Block RAM
7 series FPGAs DDR3 SDRAM ²	14,016	9,019	4 ^{3, 4}	1	2 ³	2
7 series FPGAs DDR2 SDRAM ²	9,267	6,038	2	1	1	0
7 series FPGAs QDR II+ SRAM	3,209	2,568	2	1	1	0
7 series FPGAs RLDRAM II	6,261	4,519	2	1	1	7
7 series FPGAs RLDRAM 3	9,039	7,950	2	1	1	12
7 series FPGAs LPDDR2 SDRAM	3,952	3,285	2	1	1	0

Notes:

1. Resource utilization can change depending on the options chosen, memory device used, or both. Resource information is provided for 72-bit DDR3 SDRAM, 72-bit DDR2 SDRAM, 36-bit QDR II+ SRAM, 72-bit RLDRAM II, 72-bit RLDRAM 3, and 32-bit LPDDR2 SDRAM interfaces.
2. UDIMM 72-bit designs (ECC disabled).
3. For design frequencies > 667 MHz, three BUFs and two MMCM are used in the design. For design frequencies < 667 MHz, only two BUFs and one MMCM are used in the design.
4. One BUFG used for the clock during write calibration.

Verification

The AMD 7 series FPGAs memory interface solutions cores have been verified in simulation. Verification tests include:

- Initialization sequence
- Read calibration
- Memory read operation
- Memory write operation
- Row/bank management

- Write leveling

Finding Additional Documentation

Technical Information Portal

The AMD Technical Information Portal is an online tool that provides robust search and navigation for documentation using your web browser. To access the Technical Information Portal, go to <https://docs.amd.com>.

Documentation Navigator

Documentation Navigator (DocNav) is an installed tool that provides access to AMD Adaptive Computing documents, videos, and support resources, which you can filter and search to find information. To open DocNav:

- From the AMD Vivado™ IDE, select **Help** → **Documentation and Tutorials**.
- On Windows, click the **Start** button and select **Xilinx Design Tools** → **DocNav**.
- At the Linux command prompt, enter `docnav`.

Note: For more information on DocNav, refer to the *Documentation Navigator User Guide* ([UG968](#)).

Design Hubs

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- In DocNav, click the **Design Hubs View** tab.
- Go to the [Design Hubs](#) web page.

References

These documents provide supplemental material useful with this guide:

1. JEDEC Standard JESD79-3E: DDR3 SDRAM, JEDEC Solid State Technology Association JEDEC Standard JESD79-2F: DDR2 SDRAM Specification, JEDEC Solid State Technology Association [JEDEC web page](#)

This AMD document can be located on the [MIG Solution Center Documentation page](#):

1. *Kintex 7 FPGAs Data Sheet: DC and AC Switching Characteristics* ([DS182](#))
2. *Virtex 7 FPGAs Data Sheet: DC and AC Switching Characteristics* ([DS183](#))
3. *Spartan 7 FPGAs Data Sheet: DC and AC Switching Characteristics* ([DS189](#))
4. *Zynq 7000 SoC and 7 series Devices Memory Interface Solutions* ([UG586](#))
5. *Vivado Design Suite User Guide: Designing with IP* ([UG896](#))
6. *ISE to Vivado Design Suite Migration Guide* ([UG911](#))
7. *Vivado Design Suite User Guide: Release Notes, Installation, and Licensing* ([UG973](#))

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Revision History

The following table shows the revision history for this document.

Section	Revision Summary
11/13/2024 Version 4.2	
All sections	Vivado Design Suite release for MIG v4.2.
05/30/2024 Version 4.2	
All sections	Vivado Design Suite release for MIG v4.2.
All sections	Vivado Design Suite release for MIG v4.2.
12/05/2018 Version 4.2	
All sections	Vivado Design Suite release for MIG v4.2.
04/04/2018 Version 4.1	
All sections	Vivado Design Suite release for MIG v4.1.
03/02/2018 Version 4.1	
All sections	Reverted doc version to v4.1 to match Vivado Design Suite release for MIS core v4.1.
10/04/2017 Version 4.2	
All sections	Vivado Design Suite release for MIG v4.2.
Features	Added QDR II+ density device support.
RLDRAM II Features	Added RLDRAM II density device support.
RLDRAM 3 Features	Added RLDRAM 3 density device support.
06/07/2017 Version 4.2	
All sections	Vivado Design Suite release for MIG v4.2. Added Spartan 7 support and changed it to 7 series family.
04/05/2017 Version 4.2	
All sections	Vivado Design Suite release for MIG v4.2.
11/30/2016 Version 4.1	
All sections	Vivado Design Suite release for MIG v4.1.
10/05/2016 Version 4.1	
All sections	Vivado Design Suite release for MIG v4.1.
06/08/2016 Version 4.0	
All sections	Vivado Design Suite release for MIG v4.0.
04/06/2016 Version 3.0	
All sections	Vivado Design Suite release for MIG v3.0.
11/18/2015 Version 2.4	
All sections	Vivado Design Suite release for MIG v2.4.

Section	Revision Summary
09/30/2015 Version 2.4	
All sections	Vivado Design Suite release for MIG v2.4.
AXI4 Slave Interface Features	Added 72 bits with eight bits of ECC.
06/24/2015 Version 2.3	
All sections	Vivado Design Suite release for MIG v2.3.
04/01/2015 Version 2.3	
All sections	Vivado Design Suite release for MIG v2.3.
Resource Utilization	Updated table.
11/19/2014 Version 2.3	
All sections	Vivado Design Suite release for MIG v2.3.
10/01/2014 Version 2.2	
All sections	Vivado Design Suite release for MIG v2.2.
06/04/2014 Version 2.1	
All sections	Vivado Design Suite release for MIG v2.1.
Resource Utilization	Updated table.
12/18/2013 Version 2.0	
All sections	Vivado Design Suite release for MIG v2.0. Updated doc title.
10/02/2013 Version 2.0	
All sections	Vivado Design Suite release for MIG v2.0.
IP Facts	Updated simulation.
06/19/2013 Version 2.0	
All sections	Vivado Design Suite release for MIG v2.0. Revision number advanced to 2.0 to align with core version number.
03/20/2013 Version 1.8	
All sections	ISE 14.5 and Vivado Design Suite release for MIG v1.9.
Resource Utilization	- Moved Resource table into a separate Resource section. - Updated 7 Series FPGAs DDR3 SDRAM in Resource Utilization table. Added LPDDR2 SDRAM content.
12/18/2012 Version 1.7	
All sections	ISE 14.4 and Vivado Design Suite release for MIG v1.8. Added 8 Gb to DDR3 SDRAM feature. Added VHDL support.
10/16/2012 Version 1.6	
All sections	ISE 14.3 and Vivado Design Suite release for MIG v1.7. Added RDRAM 3 content.
07/25/2012 Version 1.5	
All sections	ISE 14.2 and Vivado Design Suite release for MIG v1.6.
04/24/2012 Version 1.4	
All sections	ISE 14.1 and Vivado Design Suite release for MIG v1.5. Added VHDL source code for top-level files for all memory devices. For DDR3 and DDR2 SDRAM, added: I/O Power Reduction option, AXI4-Lite interface support for ECC control and status registers, and 72-bit data width.
01/18/2012 Version 1.3	
All sections	ISE 13.4 and Vivado Design Suite release for MIG v1.4. For DDR3 SDRAM: Added support for 4 Gb density, DDR3L (1.35 V), and dual rank UDIMM, RDIMM, and SODIMM. Removed support for AXI4-Lite interface and 72-bit data width. Added DDR2 SDRAM support.

Section	Revision Summary
10/19/2011 Version 1.2	
All sections	ISE 13.3 and Vivado Design Suite release for MIG v1.3. Added Resources to the IP Facts table. For DDR3 SDRAM, added support for up to eight controllers, added 2:1 as an interface clock ratio, added AXI4-Lite interface support, and added 72 as a memory data width option. For QDR II+ SRAM, added support for 2-word bursts and support for up to eight controllers. For RLD RAM II, added support for Address Multiplexing Mode and support for up to eight controllers.
06/22/2011 Version 1.1	
All sections	ISE 13.2 software release. Added RLD RAM II support throughout document. Added single rank UDIMM support bullet to DDR3 SDRAM Features. Added internal V_{REF} support.
03/01/2011 Version 1.0	
Initial release.	N/A

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